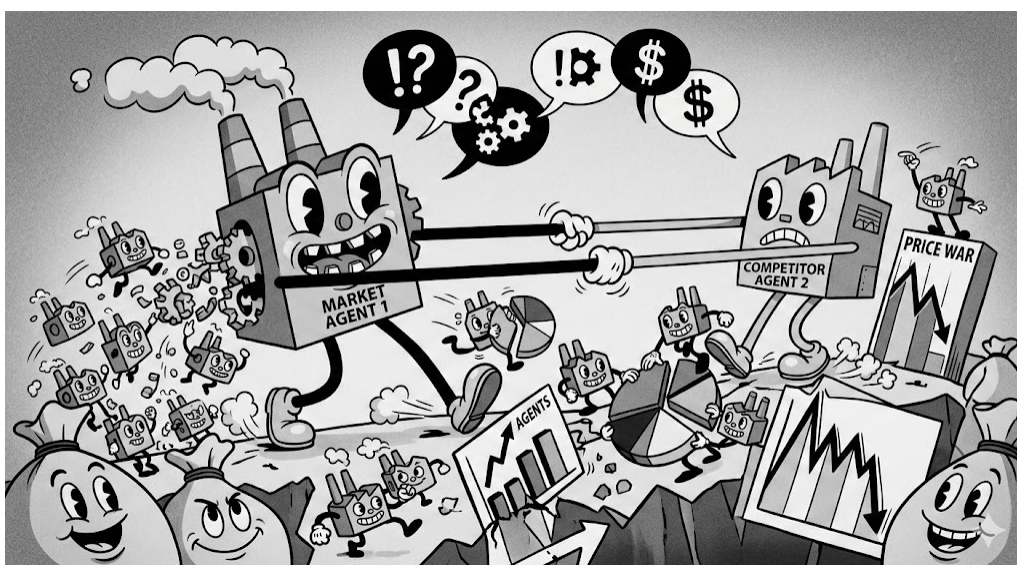


Agent-Based Modeling — Coursework 1

CASA0011 — Systematic Experimentation (SugarScape)

*Evolutionary Dynamics of Market Competition:
From Individual Survival to Corporate Expansion*



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1 Background

The SugarScape model (Epstein and Axtell, 1996) was originally designed to simulate how complex social phenomena emerge from simple interactions among heterogeneous individuals competing for limited resources. The model series can also be applied in market competition context. Park et al. (2024) adapted the SugarScape framework in a housing market to study how agents’ demand and limited vision decide market outcomes. Based on this, this study extends the model to firm-level competition.

2 Aim

This study reinterprets the SugarScape model 2 (Constant Growback) and 3 (Wealth Distribution) from the NetLogo Library (Wilensky, 1999a, 1999b) as models of firm-level market competition. In this context, I translate metabolism into operating costs, vision into market awareness, and accumulated sugar into retained capital. Then two progressive stages are designed to examine how market structure and competitive pressure jointly shape firm characteristics and market inequality. Research questions are set below:

Stage A (based on Model 2): In a static and resource-limited market, how does firm density reshape the distribution of firm characteristics through competition?

Stage B (based on Model 3): In a dynamic market with a brand expansion mechanism, how entry conditions for opening branch shape market structure through a filtering process. Does efficiency gain from enterprise scaling further concentrate wealth among dominant brands, creating a winner-takes-more market dynamic?

Together, the two stages form a progressive comparative framework from static competition to dynamic expansion.

3 Method

This study enhances two SugarScape models and conducts systematic parameter experiments with Netlogo BehaviorSpace. To ensure selection is entirely through market competition rather than lifespan randomness of the initial model, bankruptcy (retained capital falling to zero) is set as the exit mechanism for both.

To observe and compare cross-stage market resource absorption efficiency, I introduce market utilization (U) as a common benchmark in the two stages, which is defined as follows:

$$U^1 = \frac{\sum \text{operating cost of all firms}}{\sum \text{sugar regrown across all patches}} \times 100\% \quad (1)$$

3.1 Stage A: Static Competitive Market

Stage A extends the SugarScape 2 Constant Growback model with two additional reporters: market utilization and Gini Index² (measuring wealth distribution). The initial number of firms N (market density) is the sole independent variable. Observed metrics are mean operating cost, mean market reach, market utilization, and Gini Index. Experimental parameters are detailed in Table 1.

¹In the Stage B secondary experiment, U exceeding 100% is because the introduction of the same-brand adjacency efficiency mechanism (b) improves firms’ actual harvesting efficiency.

²The Gini Index measures wealth inequality within a population, ranging from 0 (perfect equality, all firms hold equal wealth) to 1 (perfect inequality, one firm holds all wealth). A higher value indicates greater wealth concentration.

3.2 Stage B: Dynamic Expansion Market

The original SugarScape 3 reproduction mechanism replaces each dying agent with a single offspring in the same location, modelling biological succession rather than organisational growth. Stage B replaces it with a brand expansion mechanism, where wealthy firms open new branches within their own market reach. The new branding mechanism comprises four rules:

1. When a firm's wealth exceeds the expansion threshold ET, it attempts to open a new branch each tick with probability p .
2. The parent firm transfers a random amount drawn from $[MW, ET]$ (where MW is the minimum entry wealth and ET is the expansion threshold) to the new branch, reducing its own wealth accordingly.
3. The new branch inherits the parent's brand identity, draws operating cost and market reach independently from the same initial distributions, and is placed on a random vacant patch within the parent's market reach (vision) range.
4. If a firm has at least one same-brand neighbour (up/down/left/right), its actual harvested is $S'_i = S_i \times (1 + b)$, where b is the adjacent harvest bonus and $b = 0$ represents the no-scale-effect baseline.

With brand feature introduced, the only wealth distribution measurement Gini Index is not enough. Brand-level Gini reporter is added to the model, forming a dual-layer inequality measurement system.

The experiment is conducted through two sub-stages. The primary experiment adopts a full factorial design across ET and MW to examine the interaction effects of entry conditions. The secondary experiment fixes ET and MW and sweeps Adjacency Harvest Bonus (b) to isolate the effect of scale efficiency on market structure. Observed metrics across both experiments are mean operating cost, mean market reach, market utilization, individual-level Gini Index, and brand-level Gini Index. Full parameters are detailed in Table 1.

3.3 Stages Experiments Parameter Summary

Table 1: Experiment Parameter Summary

Stage	Parameter(s)	Values Tested	Fixed Parameters
<i>Universal constants: operating cost $\sim U(1, 4)$, market reach $\sim U(1, 6)$, sugar growback = 1</i>			
A — Model 2	Initial number of firms N (market density)	10–2499 (15 levels, density-stratified sampling)	initial capital $\sim U(5, 25)$
B — Model 3 (primary)	Expansion threshold (ET) $\times$ Minimum entry wealth (MW)	ET: {10, 20, 30, 40, 50}; MW: {0, 9, 19, 29, 39, 49}; ET > MW (20 valid combinations)	$N = 400, p = 0.6, b = 0$
B — Model 3 (secondary)	Adjacency harvest bonus b	0–1.0 (step = 0.1)	$N = 400, p = 0.6, ET = 20, MW = 10$

Note: All experiments were repeated 30 times. Stage A ran for 500 steps and Stage B for 1000 steps; pre-experiments confirm all parameter combinations reach steady state within these limits. Stage A adopts density-stratified sampling: $N \in \{10, 50, 100, 200, 300, 400, 600, 800, 1000, 1200, 1400, 1600, 1800, 2000, 2200, 2499\}$, with finer intervals at low density ($N < 200$) to capture early competitive selection dynamics, and $N = 2499$ included to observe market behaviour at the firm population ceiling. MW is designed as {0, 9, 19, 29, 39, 49} so that each ET level includes an $MW = ET - 1$ case, capturing the effect of a near-zero startup capital range. Branching probability $p = 0.6$ is validated by pre-experiments to ensure market convergence within the time limit and treated as a control parameter.

4 Results

4.1 Stage A

Figure 1: Evolution of Firm Characteristics Under Market Competition (Stage A)

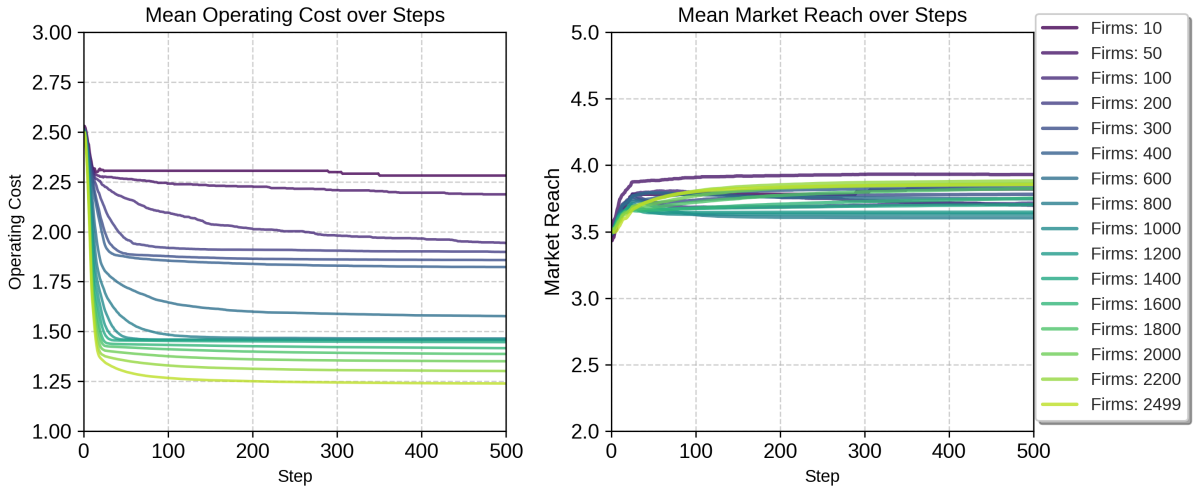


Figure 1 shows mean operating cost declining nonlinearly from approximately 2.3 to 1.25 as N increases. Mean market reach decreases first and then slightly increases, reaching its minimum when $N = 800$. Figure 2 compares standard deviation bands at three density levels. At $N = 10$, the range of dispersion in operating costs and market reach is extremely wide. As market density increases, both metrics become

stable.

Figure 2: Evolution of Firm Traits by Market Density (Stage A)

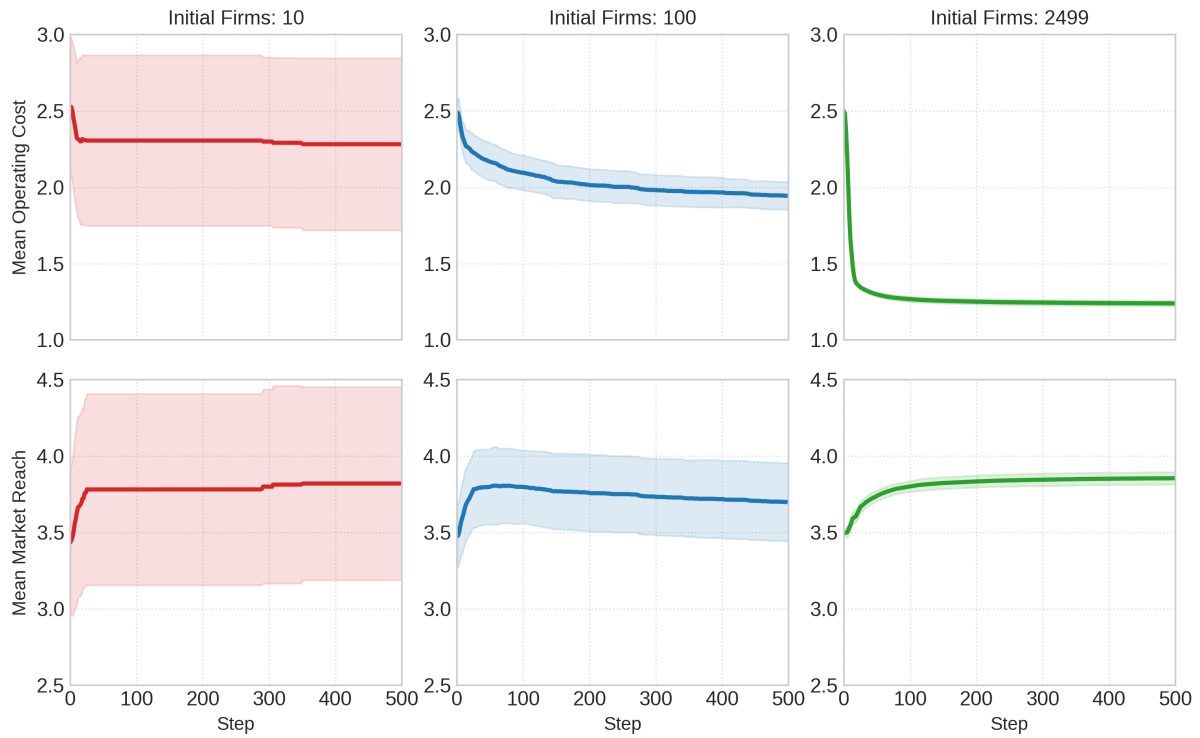
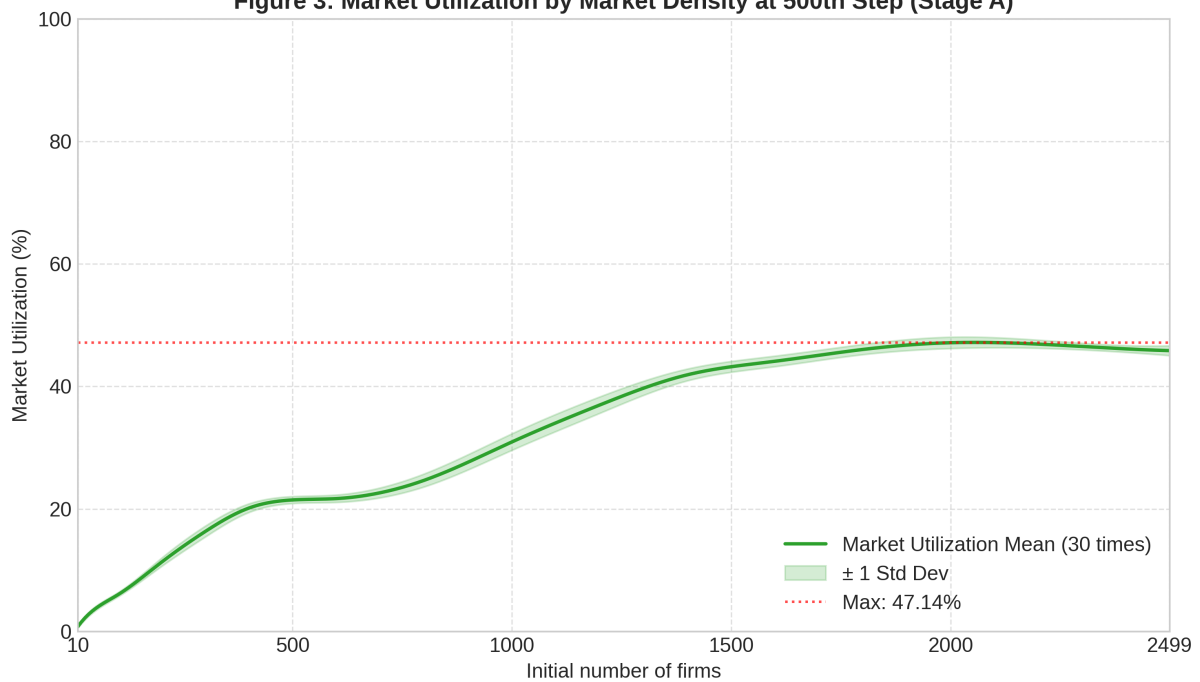


Figure 3 shows market utilization rising continuously with N , reaching 47.14% at $N = 2000$. Utilization fails to increase further even at the model ceiling of $N = 2499$, indicating a structural ceiling on resource absorption.

Figure 3: Market Utilization by Market Density at 500th Step (Stage A)



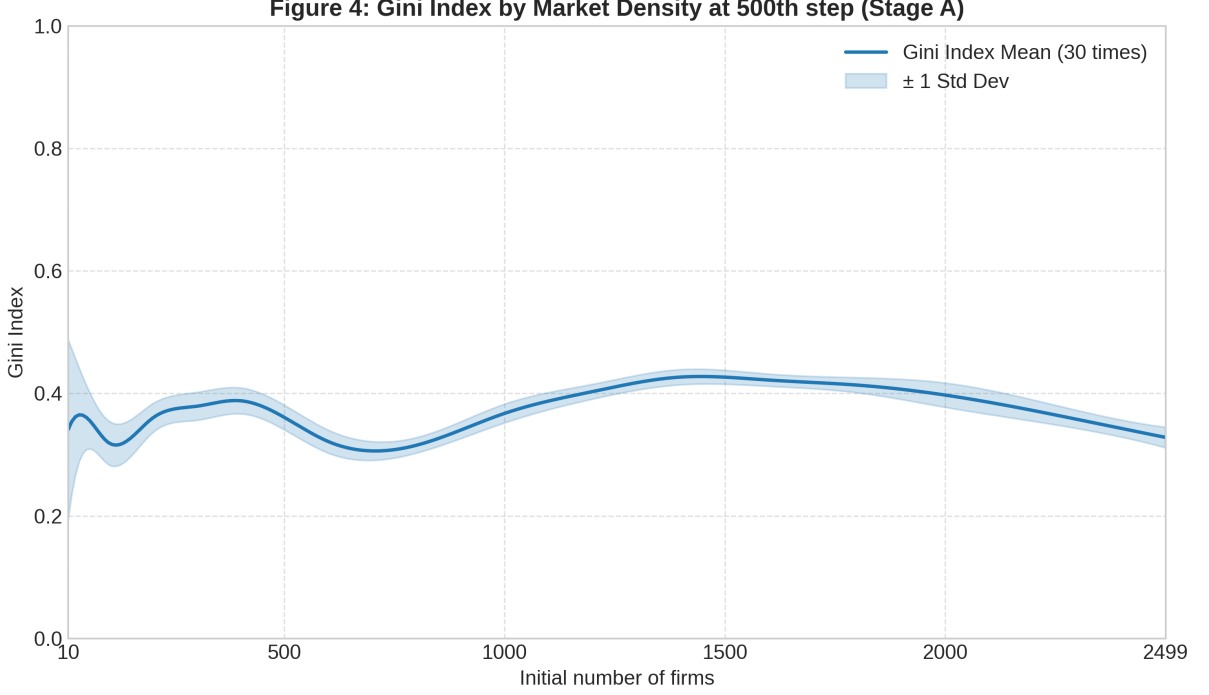


Figure 4 shows the Gini Index following a non-monotonic pattern. When $N < 100$, the standard deviation range is wide, reflecting highly random outcomes. The Gini then declines to approximately 0.30 at $N = 800$, peaks at 0.43 around $N \approx 1400$, and falls to approximately 0.33 at higher densities.

When $N = 800$, resource-rich patches begin to saturate, marking a transition from cost-based to location-based selection. Below this threshold, competitive pressure is insufficient to eliminate high-cost firms; above it, positional advantage dominates survival outcomes. This explains why $N = 800$ coincides with the minimum values of both the Gini Index and mean market reach.

4.2 Stage B — Primary Experiment ($ET \times MW$)

Table 2: Mean Market Reach (Vision) by Entry Conditions ($ET \times MW$), mean $\pm$ std

$ET \setminus MW$	0	9	19	29	39	49
10	3.50 ± 0.04	3.50 ± 0.05	—	—	—	—
20	3.50 ± 0.06	3.50 ± 0.04	3.51 ± 0.04	—	—	—
30	3.51 ± 0.05	3.50 ± 0.05	3.48 ± 0.06	3.52 ± 0.05	—	—
40	3.52 ± 0.05	3.51 ± 0.04	3.50 ± 0.04	3.51 ± 0.04	3.52 ± 0.03	—
50	3.51 ± 0.03	3.50 ± 0.04	3.49 ± 0.04	3.49 ± 0.03	3.53 ± 0.05	3.52 ± 0.04

Note: Cells marked “—” represent invalid combinations where $ET \leq MW$. All values rounded to 2 decimal places.

Table 3: Mean Operating Cost (Metabolism) by Entry Conditions (ET $\times$ MW), mean $\pm$ std

ET \ MW	0	9	19	29	39	49
10	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	—	—	—	—
20	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.001	—	—	—
30	1.000 $\pm$ 0.000	1.001 $\pm$ 0.001	1.001 $\pm$ 0.001	1.004 $\pm$ 0.002	—	—
40	1.001 $\pm$ 0.001	1.003 $\pm$ 0.002	1.006 $\pm$ 0.003	1.009 $\pm$ 0.003	1.018 $\pm$ 0.005	—
50	1.004 $\pm$ 0.002	1.006 $\pm$ 0.003	1.015 $\pm$ 0.004	1.023 $\pm$ 0.006	1.034 $\pm$ 0.005	1.044 $\pm$ 0.007

Note: Cells marked “—” represent invalid combinations where $ET \leq MW$. Bold indicates the maximum observed deviation from the model floor. All values rounded to 3 decimal places.

Tables 2 and 3 show mean market reach stable at 3.50 ± 0.05 and mean operating cost converging to the model floor of 1.00 under most conditions.

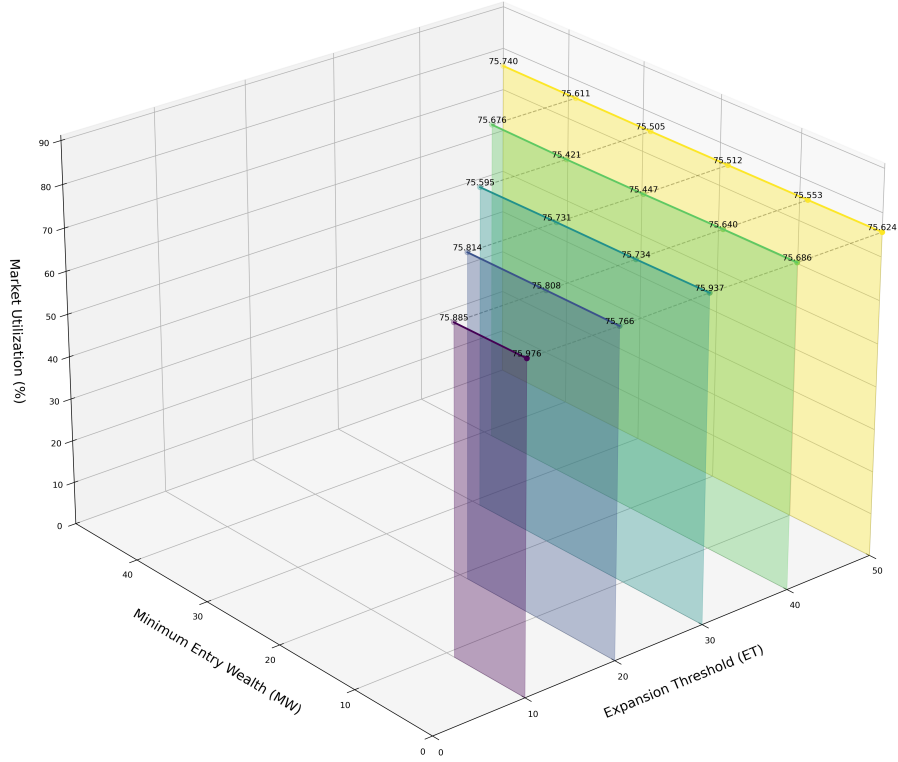
Figure 5: Market Utilization by Entry Conditions (ET $\times$ MW) (Stage B-Primary)

Figure 5 shows market utilization is similarly insensitive to ET and MW, staying between 75.4% and 76.0%.

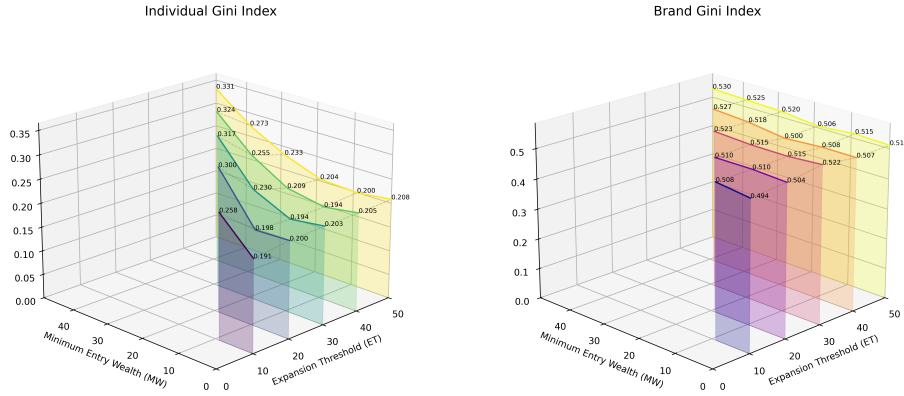
Figure 6: Individual Gini vs Brand Gini under Entry Conditions (ET × MW) (Stage B-Primary)

Figure 6 shows that individual Gini rises with MW (minimum entry wealth) and broadly declines with ET (expansion threshold). When the entry capital range [MW, ET] narrows sharply, inequality intensifies further, peaking at 0.331 when $ET = 50$ and $MW = 49$. Brand Gini remains highly stable across all 20 valid combinations, ranging from 0.494 to 0.530.

4.3 Stage B — Secondary Experiment (Adjacency Harvest Bonus)

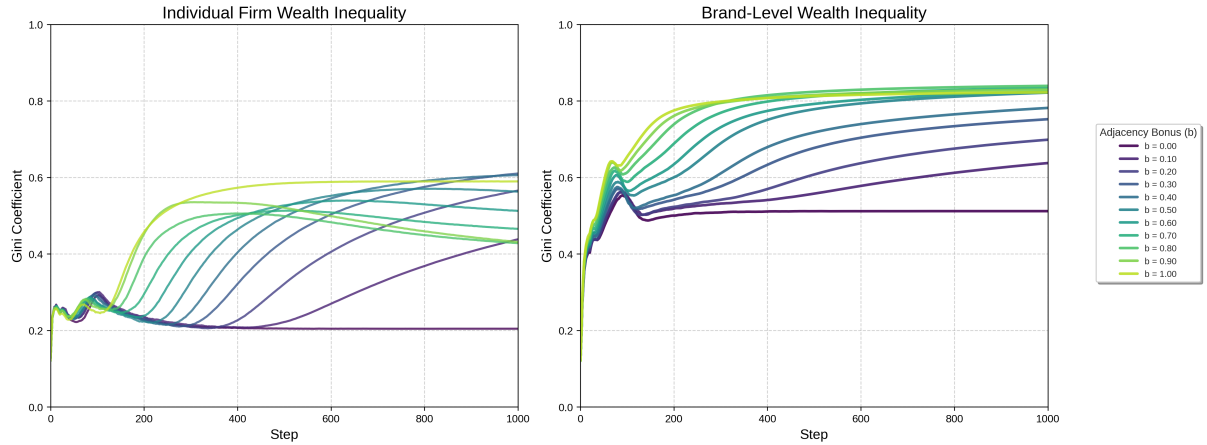
Figure 7: Impact of Adjacency Harvest Bonus(b) on Market Inequality Dynamics (Stage B-Secondary)

Figure 7 reveals contrasting patterns across the two Gini measures. Individual Gini rises from 0.205 at $b = 0$ to a peak of 0.610 at $b = 0.3$. It then declines to 0.428 at $b = 0.8$, before jumping again to 0.590 at $b = 1.0$. Brand Gini behaves differently. It jumps immediately from 0.512 to above 0.638 once $b > 0$, and continues rising to a peak of 0.839 when $b = 0.8$.

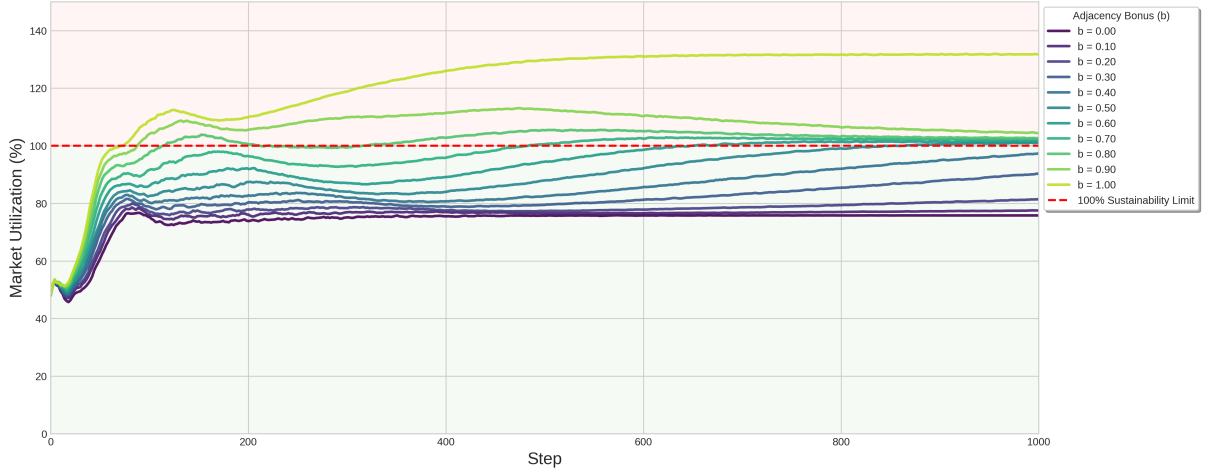
Figure 8: Market Utilization by Adjacency Bonus (b) (Stage B-Secondary)

Figure 8 shows market utilization rising steadily with b . It first exceeds 100% at $b = 0.5$, reaching 131.7% when $b = 1.0$.

5 Discussion

5.1 Asymmetric Selection: Cost Over Awareness

Both stages show that operating cost declines significantly under competitive pressure, while market reach remains relatively stable, indicating that high-cost firms are the first to go bankrupt and positional advantages are weakened as demand-rich patches become occupied in dense markets.

5.2 Structural Changes in Market Utilization

Stage A's utilization ceiling of approximately 47% is independent of firm count, suggesting it is determined by the spatial distribution of market demand and the rate of demand renewal (sugar regeneration rate) in static market. In the Stage B primary experiment, the expansion mechanism stabilises utilization at 75.5%. when the adjacent harvest bonus is introduced, utilization exceeds 100% and ultimately reaches 131.7% as bonus rate increases. These jointly confirm a more scaled firm composition will drive expansion of market carrying capacity.

5.3 Decoupling of Two Forms of Inequality

In the Stage B primary experiment, when ET and MW are close in value, parent companies give homogeneous money for new firms as their startup capital. Thus the competition produces more polarised results at the aggregate level. Conversely, the wider capital range introduces heterogeneity, reducing individual-level inequality. Brand Gini remains approximately 0.51, confirming that although the expansion mechanism reduces individual-level inequality, it introduces a higher level of inequality at the brand level.

5.4 The Cross-Stage Decisive Role of Pre-Entry Market Intelligence

The wide standard deviation bands at low density in Stage A indicate the importance of occupying resource-rich locations upon entry. It exerts significantly greater influence on individual outcomes than operational traits. In Stage B, the transmission of geographic advantage across generations enables dominant brands to accumulate wealth rapidly and becomes a structural source of brand-level inequality.

The adjacency harvest bonus further amplifies this mechanism. Economies of scale raise clusters' efficiency, and a positive feedback loop pushes brand Gini to its extreme peak of 0.839.

6 Conclusion

This study examines how competitive pressure and market structure jointly shape firm survival and wealth inequality across stages. In the static market, the selection mechanism eliminates high-cost firms first, not those with weaker market awareness. In the dynamic expansion market, entry capital conditions: expansion threshold and minimum entry wealth affect individual-level inequality through the startup capital range. Brand-level wealth divergence, however, originates from random geographic positioning upon entry and compounds across generations as branches inherit locational advantage, which entry capital cannot offset. When the adjacency harvest bonus is introduced, scale efficiency further raises market carrying capacity and reinforces the positive feedback loop of geographic first-mover advantage, driving extreme brand-level inequality. The results also suggest that for firms and brands cost control is determinant of survival and intelligence in occupying demand-rich locations upon entry is fundamental to success.

References

- [1] Epstein, J. M. and Axtell, R. (1996) *Growing Artificial Societies: Social Science from the Bottom Up*. MIT Press.
- [2] Wilensky, U. (1999a) *NetLogo Models Library: Sugarscape 2 Constant Growback*. Evanston, IL: Center for Connected Learning and Computer-Based Modeling, Northwestern University. Available at: <https://ccl.northwestern.edu/netlogo/models/Sugarscape2ConstantGrowback>
- [3] Wilensky, U. (1999b) *NetLogo Models Library: Sugarscape 3 Wealth Distribution*. Evanston, IL: Center for Connected Learning and Computer-Based Modeling, Northwestern University. Available at: <https://ccl.northwestern.edu/netlogo/models/Sugarscape3WealthDistribution>
- [4] Park, D., Hong, J. and Ryu, D. (2024) ‘Heterogeneous expectations in the housing market: a sugarscape agent-based model’, *Journal of Housing and the Built Environment*, 39, pp. 1465–1489.

Appendix

A. Stage A: Extended Code Chunk Based on Sugarscape Model 2

The complete model of **Stage-A: Static-Competitive-Market**, can be found in the following repository:

https://github.com/zhengpei-xu/CASA/blob/main/CASA0011_Agent-based-Modelling-for-Spatial-Systems/Coursework-1_Analysing-Models/Models/Stage-A_Static-Competitive-Market.nlogo

Listing 1: Reporter for market-utilization

```

1  ;; -----addition part 1-----
2  ;; -----reportor for market-utilization-----
3  to-report get-market-utilization
4    ;; 1. calculating the total metabolism of agents (total cost every tick)
5    let total-demand sum [metabolism] of turtles
6
7    ;; 2. caculating total Theoretical Maximum output (potential total market
      output)
8    ; because the potential recovery value of each sugar mine is 1/tick
9    ; now can use the number of mines to represent potential total market output
      (2069)
10   let total-supply count patches with [max-psugar > 0]
11
12   ; solving Zero Division Error
13   if total-supply = 0 [report 0]
14
15   ;; 3. reporting percentage
16   report (total-demand / total-supply) * 100
17
18 end

```

Listing 2: Reporter for Gini Index

```

1  ;; -----addition part 2-----
2  ;; -----plot for Gini Index for wealth distribution-----
3  to update-lorenz-and-gini
4    let num-people count turtles
5    let sorted-wealths sort [sugar] of turtles
6    let total-wealth sum sorted-wealths
7    let wealth-sum-so-far 0
8    let index 0
9    set gini-index-reserve 0
10   set lorenz-points []
11   repeat num-people [
12     set wealth-sum-so-far (wealth-sum-so-far + item index sorted-wealths)
13     set lorenz-points lput ((wealth-sum-so-far / total-wealth) * 100) lorenz-
      points
14     set index (index + 1)
15     set gini-index-reserve
      gini-index-reserve +
16       (index / num-people) -
17       (wealth-sum-so-far / total-wealth)
18   ]
19
20 end

```

Listing 3: Optional addition for brands property

```

1 ;; -----addition part 3-----
2 ;; -----adding brands property-----
3 ;; -----this is optional-----
4 ;;-----just for aesthetic purposes-----
5 to distribute-brands
6   ;; Create the initial population based on the UI slider
7   create-turtles initial-population [
8     ;; Assign the unique 'who' number as the Brand ID
9     set brand who
10
11     ;; Set a unique color based on the Brand ID to visualize diversity
12     ;; The modulo operator ensures colors stay within the NetLogo visible range
13     .
14     ;; Gemini used here to assign color
15     set color (5 + (brand * 13) mod 135)
16
17     ;; Call the general turtle setup for traits and positioning
18     turtle-setup
19   ]
20 end

```

B. Stage B: Extended Code Chunk Based on Sugarscape Model 3

The complete model of **Stage B: Dynamic Expansion Market**, can be found in the following repository:

https://github.com/zhengpei-xu/CASA/blob/main/CASA0011_Agent-based-Modelling-for-Spatial-Systems/Coursework-1_Analysing-Models/Models/Stage-B_Dynamic-Expansion-Market.nlogox

Listing 4: Introduction of brands property

```

1 ;; -----addition part 1-----
2 ;; -----adding brands property-----
3 to distribute-brands
4   ;; Create the initial population based on the UI slider
5   create-turtles initial-population [
6     ;; Assign the unique 'who' number as the Brand ID
7     set brand who
8
9     ;; Set a unique color based on the Brand ID to visualize diversity
10    ;; The modulo operator ensures colors stay within the NetLogo visible range
11    .
12    ;; Gemini used here to assign color
13    set color (5 + (brand * 13) mod 135)
14
15    ;; Call the general turtle setup for traits and positioning
16    turtle-setup
17  ]
18 end

```

Listing 5: Introduction of new expansion mechanism

```

1 ;; -----addition part 2-----

```

```

2  ;; -----introducing new expansion mechanism-----
3  to handle-life-cycle ;; turtle procedure
4    ;; 1. Initialize Expansion Cost
5    let birth-cost random-in-range minimum-sugar-endowment maximum-sugar-
      endowment
6
7    ;; --- 2. ACTIVE EXPANSION ---
8    ;; Prudence Check!!! Ensure expansion doesn't bankrupt the parent
9    if (sugar > maximum-sugar-endowment) and (not first-tick?) [
10
11      ;; Safety Buffer: Assets after cost must exceed 1 tick of metabolism
12      if (sugar - birth-cost > metabolism) [
13        let target-patch one-of neighbors4 with [not any? turtles-here]
14
15        if (random-float 1.0 < birth-rate) and (target-patch != nobody) [
16          set sugar (sugar - birth-cost)
17
18          hatch 1 [
19            ;; inherit 'brand', but reset capabilities
20            turtle-setup
21            set sugar birth-cost
22            ; first-tick protection buff
23            set first-tick? true
24            move-to target-patch
25          ]
26        ]
27      ]
28    ]
29
30    ;; --- 3. STARVATION CHECK (Final Check) ---
31    if sugar <= 0 [
32      die
33    ]
34
35    if first-tick? [ set first-tick? false ]
36  end

```

Listing 6: Addition of adjacency harvest bonus

```

1  ;; -----addition part 3-----
2  ;; -----adding adjacency harvest bonus-----
3  to turtle-eat ;; turtle procedure
4    ;; 1. Check for same-brand neighbours in the four cardinal directions (up/
      down/left/right)
5    let my-brand brand
6    let allies-nearby (turtles-on neighbors4) with [brand = my-brand]
7
8    ;; 2. Determine if a bonus should be applied
9    let collection-amount psugar
10
11    if any? allies-nearby [
12      ;; Applying a bonus if at least one same-brand neighbour is directly
        adjacent
13      set collection-amount (psugar * (1 + efficiency-gain))
14    ]
15

```

```

16 ;; 3. Update sugar storage (subtract metabolism + add boosted collection)
17 set sugar (sugar - metabolism + collection-amount)
18
19 ;; 4. Set patch sugar to 0 after consumption
20 set psugar 0
21 end

```

Listing 7: Reporter for market-utilization

```

1 ;; -----addition part 4-----
2 ;; -----reporter for market-utilization-----
3 to-report get-market-utilization
4 ;; 1. calculating the total metabolism of agents (total cost every tick)
5 let total-demand sum [metabolism] of turtles
6
7 ;; 2. caculating total Theoretical Maximum output (potential total market
8   output)
9 ; because the potential recovery value of each sugar mine is 1/tick
10 ; now can use the number of mines to represent potential total market output
11   (2069)
12 let total-supply count patches with [max-psugar > 0]
13
14 ; solving Zero Division Error
15 if total-supply = 0 [report 0]
16
17 ;; 3. reporting percentage
18 report (total-demand / total-supply) * 100
19 end

```

Listing 8: Reporter for Brand level Gini Index

```

1 ;; -----addition part 5-----
2 ;; ----plot for Brand Gini Index for brand inequality----
3 to update-brand-lorenz-and-gini
4 let alive-brand-ids remove-duplicates [brand] of turtles
5 let brand-wealths map [ b -> sum [sugar] of turtles with [brand = b] ] alive-
6   brand-ids
7 set num-brands length brand-wealths
8
9
10 let sorted-wealths sort brand-wealths
11 let total-wealth sum sorted-wealths
12
13 if total-wealth = 0 [ stop ]
14
15 let wealth-sum-so-far 0
16 let index 0
17 set brand-gini-index-reserve 0
18 set brand-lorenz-points []
19
20 repeat num-brands [
21   set wealth-sum-so-far (wealth-sum-so-far + item index sorted-wealths)
22   ;; Updating brand-specific lorenz list

```

```
23   set brand-lorenz-points lput ((wealth-sum-so-far / total-wealth) * 100)
      brand-lorenz-points
24   set index (index + 1)
25   ;; Calculating the specific reserve for brands
26   set brand-gini-index-reserve
      brand-gini-index-reserve +
27     (index / num-brands) -
28     (wealth-sum-so-far / total-wealth)
29
30 ]
31 end
```